ECE 501 - DIGITAL IMAGE PROCESSING

Explainable Multi-Label Content-Based Image Retrieval for Aerial Images using Classical Computer Vision

Weekly Progress Report

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1 Introduction

During Week 7, the team completed the second half of Phase 5: uncertainty estimation (Phase 5C), the final performance evaluation (Phase 5D) and the integration of all components into one retrieval system. This work uses the tuned fusion weights and the explanations produced in Week 6. With this week's work, all phases of the CBIR project are implemented.

2 Review of Week 6

In Week 6, the fusion weights of the 11 descriptors were tuned on a validation split, which raised same-class Precision@5 on the held-out queries from 50.10% to 56.50%. Per-cue score breakdowns and region heatmaps were also produced. The final ranking of a query is therefore known, but every result is still presented as equally trustworthy. Week 7 addresses this and measures the complete system on the held-out queries.

3 Phase 5C: Uncertainty Estimation

3.1 Indicators

Two groups of indicators are computed for the top 5 results of each query. They use only the similarity scores and never the labels. The score-margin indicators are the difference between the first and second result, the difference between the fifth result and the first one not retrieved, the difference between a result and the next one in the ranking, and how far the score stands above the score distribution of the whole gallery. The cue-disagreement indicators are the weighted spread of the gallery percentiles of a result over the 11 cues, the lowest cue percentile, and the overlap between each cue's own top 5 and the fused top 5.

3.2 Reliability model and flags

A small logistic model written in NumPy turns the indicators into an estimated reliability, once for single results (relevant means same scene class) and once for whole queries (Precision@5 of at least 0.4). It is fitted on 1,800 gallery images that were never used to tune the fusion weights, each queried against the rest of the gallery, so no test label is used for fitting. Gallery images are much easier queries than the held-out test queries (equal-weight Precision@5 is 61.5% against 50.1%), so a threshold taken from the gallery queries would flag too few test results. The flag threshold is therefore the 30th percentile of the reliability scores of the unlabelled test queries, which uses no test labels. A first version that used the threshold maximising Youden's J flagged about 61% of results and 66% of queries, which is too many to work as a warning, so it was replaced. Reason tags are added to each flag, for example cue conflict, weak score and tie with the next result.

3.3 Results

On the held-out queries (Table 1), flagged results have 37.9% same-class precision against 64.5% for unflagged results. At query level the reliability model reaches an AUC of 0.78; flagged queries have 31.4% Precision@5 against 67.2% for unflagged ones. If the system answers only the 30% of queries it rates most reliable, Precision@5 rises from 56.5% to 84.6% (Figure 1). Across the 30 classes, the mean predicted reliability follows the actual Precision@5 with a correlation of 0.72.

Some limits were found. At result level the model (AUC 0.713) is no better than the fused score alone (AUC 0.712), and the margin and cue-disagreement indicators are weak on their own (AUC 0.52 to 0.54). Their main use is to explain why a result is doubtful: results where the cues disagree on the top 5 have 27.8% precision against 63.5% otherwise. The flags follow scene-class correctness and not multi-label relevance, since at Jaccard ≥ 0.5 flagged results are not worse than unflagged ones. The predicted reliabilities are also over-confident on the test queries (mean 0.66

predicted against 0.57 observed precision), because the test queries are harder than the calibration queries: they rank results correctly but should not be read as probabilities.

Table 1: Reliability flags on the 600 held-out test queries.

Level	Share flagged	Precision flagged	Precision unflagged	All
Results (same class)	30.0%	37.9%	64.5%	56.5%
Queries (P@5)	30.0%	31.4%	67.2%	56.5%

4 Phase 5D: Performance Evaluation

4.1 Protocol

The complete system is evaluated on the 600 held-out query images against the 2,400-image gallery. The fusion weights were tuned on the validation split and the reliability model was fitted on non-tuned gallery queries, so no test label is used for fitting. The metrics are Precision@K for K of 1, 5, 10 and 20, Hit@5 and mAP over the full ranking. Relevance is defined by the scene class and by multi-label Jaccard similarity thresholds of 0.5, 0.6, 0.7 and 0.8. The final system is compared with the Phase 4 system, with each single cue, and with the equal-weight and tuned variants. A paired bootstrap with 1,000 resamples of the queries gives confidence intervals for the gains.

4.2 Results

The final system is better than the Phase 4 system at every relevance definition (Table 2 and Figure 2), and every bootstrap interval for the gain in P@5 and mAP excludes zero. The largest gain is for same-class P@5 (+6.40 points, 95% interval 4.73 to 8.07). The gains for the multi-label definitions are smaller (+1.6 to +2.6 points in P@5), partly because Precision@5 at Jaccard ≥ 0.5 is already above 84% for the Phase 4 system. These multi-label values should be read against the chance level, since a random ranking already gives 50.0% Precision@5 at Jaccard ≥ 0.5 and 13.8% at Jaccard ≥ 0.8, compared with 3.3% for the same-class definition. In the per-class results, 24 of the 30 classes improve and 5 get worse, with the largest gains for Stadium and Park (+19 points) and the largest loss for StorageTanks (−14 points). With only 20 test queries per class, the change is statistically clear for 17 classes that improve and for 3 that get worse (Airport, Beach and StorageTanks). A single weight vector for all classes is a compromise, so a few classes lose. The absolute accuracy of the handcrafted pipeline remains modest (56.5% same-class P@5, 29.2% mAP), because several scene types are visually similar.

Table 2: Phase 4 system compared with the final system on the 600 test queries (%); gains in points with 95% bootstrap intervals.

Relevance	P@5 Phase 4	P@5 final	P@5 gain [95% CI]	mAP Phase 4	mAP final	mAP gain [95% CI]
Same class	50.10	56.50	+6.40 [4.73, 8.07]	25.55	29.24	+3.69 [2.91, 4.49]
Jaccard ≥ 0.5	84.67	86.23	+1.57 [0.47, 2.73]	64.55	66.73	+2.18 [1.61, 2.77]
Jaccard ≥ 0.6	71.33	73.60	+2.27 [0.97, 3.63]	50.82	54.02	+3.20 [2.67, 3.74]
Jaccard ≥ 0.7	58.17	60.80	+2.63 [1.13, 4.20]	38.08	41.28	+3.20 [2.70, 3.73]
Jaccard ≥ 0.8	43.20	45.43	+2.23 [0.73, 3.77]	26.18	28.53	+2.34 [1.81, 2.85]

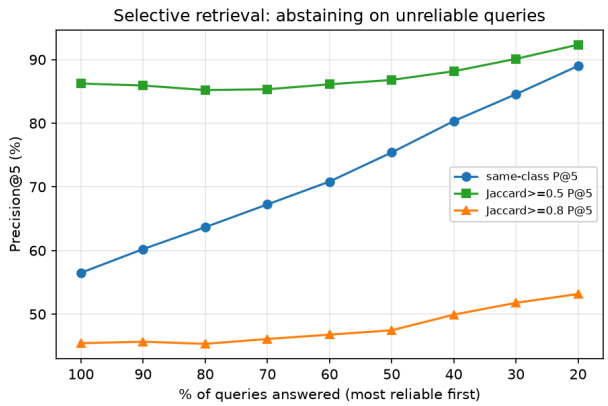


Figure 1: Precision@5 when only the most reliable queries are answered.

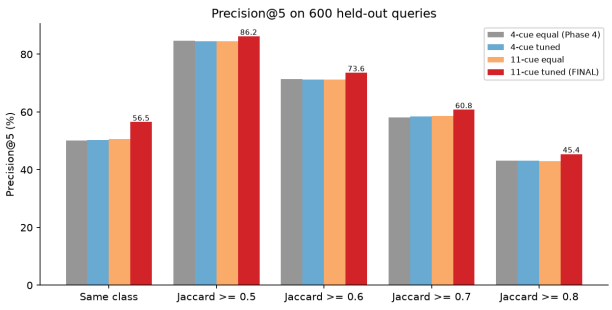


Figure 2: Precision@5 for the Phase 4, tuned and final systems at each relevance definition.

5 Final System Integration

The components were combined in one program. For a query it returns the top 5 images with the fused similarity score, the score and percentile of each cue, the region heatmaps, and a reliability estimate with flags and reasons for each result and for the query. The program was run for one representative query of each of the 30 classes. For example, the query airport_85 is difficult, since only one of its top 5 results is an airport. The system rates the query with a reliability of 63% and flags it, flags one of the five results (the terminal roof), and its explanation shows that the wrong matches (storage tanks and a terminal roof) win on LBP texture and HOG statistics and not on scene content.

6 Work Completed

During Week 7, the following work was carried out:

- Implemented the score-margin and cue-disagreement indicators and the NumPy logistic reliability model fitted on non-tuned gallery queries.
- Replaced the first flag threshold, which flagged about 60% of the results, by a 30% flag budget, added reason tags, and refitted the model on 1,800 non-tuned gallery queries with label-free thresholds after finding that gallery queries are easier than test queries.
- Evaluated the flags on the held-out queries, including the selective retrieval curve and the per-class reliability.
- Evaluated the final system with Precision@K, Hit@5 and mAP at Jaccard thresholds of 0.5 to 0.8 and for the scene class.
- Computed paired bootstrap confidence intervals and the per-class comparison with the Phase 4 system.
- Integrated ranking, explanations and flags in one final system and generated outputs for all 30 classes.
- Prepared the phase README files, summary charts and the pipeline diagram.

7 Plan for Next Week

- Update the project report and presentation with the Phase 5 results and the pipeline diagram.
- Prepare the demonstration of the final system on selected queries, including a difficult query with flags.
- Summarise the limitations: modest accuracy, classes that lose, and flags that follow scene class.

8 Conclusion

Week 7 completed Phases 5C and 5D and the final system. The reliability flags are useful at query level, since flagged queries have about half the precision of the others, but at result level they add little beyond the score itself. On the held-out queries the final system improves on the Phase 4 system at every relevance definition, with a gain of 6.4 points in same-class Precision@5. All planned phases of the CBIR project are now implemented and evaluated using classical computer vision and handcrafted features.